



Multivariate Data Analysis

Essential Prerequisites

• Probability Theory •

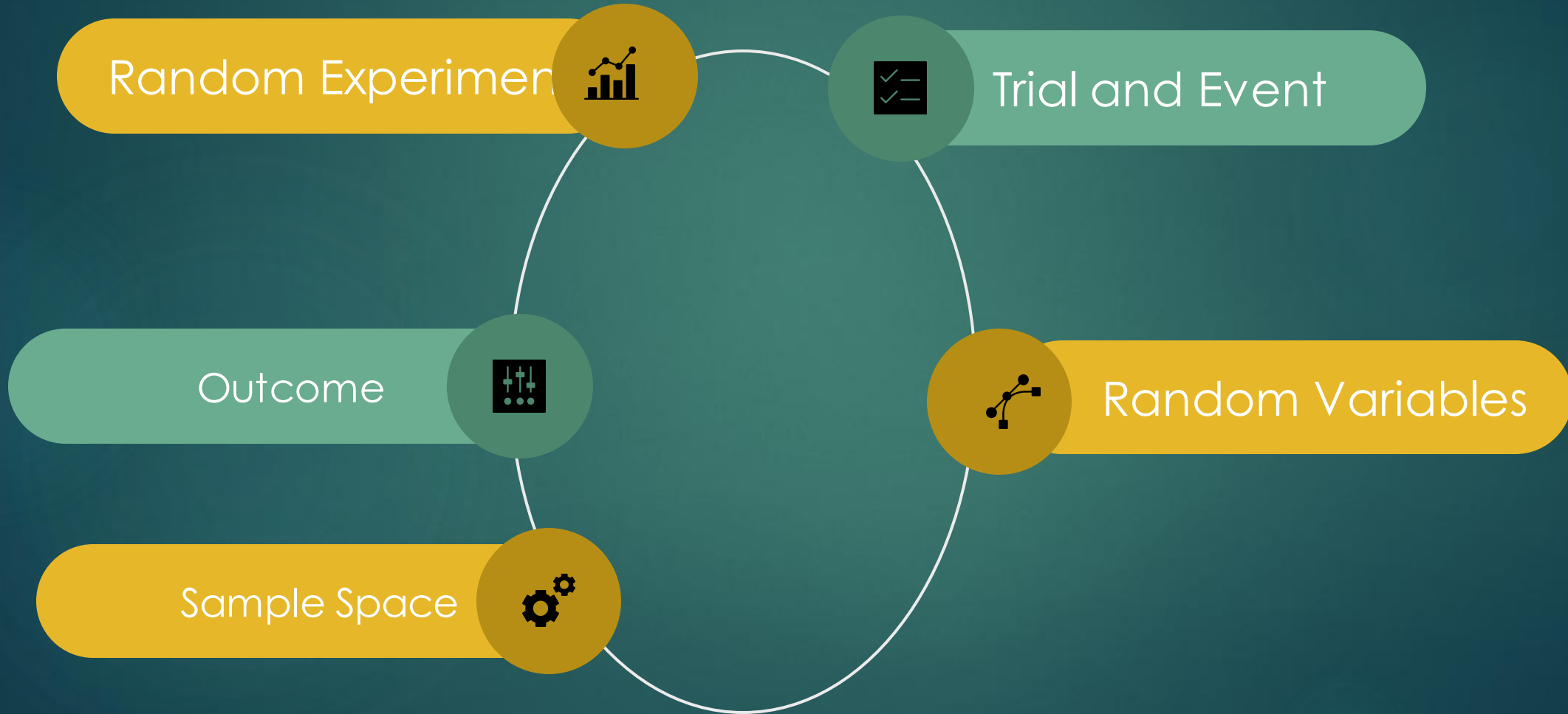
Probability is a measure of the likelihood of an event occurring.

$$\text{Probability} = \frac{\text{Number of Favourable Outcome}}{\text{Total number of Favourable Outcomes}}$$

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Random Experiment



An experiment conducted under identical conditions where the outcome is not unique, but may be any one of the possible outcomes.

Trial and Event



A trial is a single performance of an experiment, while an event is a specific outcome or set of outcomes from that trial. Essentially, a trial is the action, and the event is the result.

Outcome



The result of a random experiment.

Random Variables



Variables whose values are outcomes of a random phenomenon.

Can be discrete (e.g., number of heads) or continuous (e.g., height).

Sample Space



The set of all possible outcomes



Independent Events

The occurrence of one event does not affect the probability of the other event occurring.



Equally likely Events

Events as each outcome of an event has the same probability of occurring



Mutually Exclusive Events

Events that cannot occur at the same time. If one event happens, the other (or others) cannot happen. They are also sometimes called disjoint events.

Statistical Concepts Hierarchy

Correlation

Strength and direction of relationships

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Covariance

Direction of variable relationships

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$$

Variance

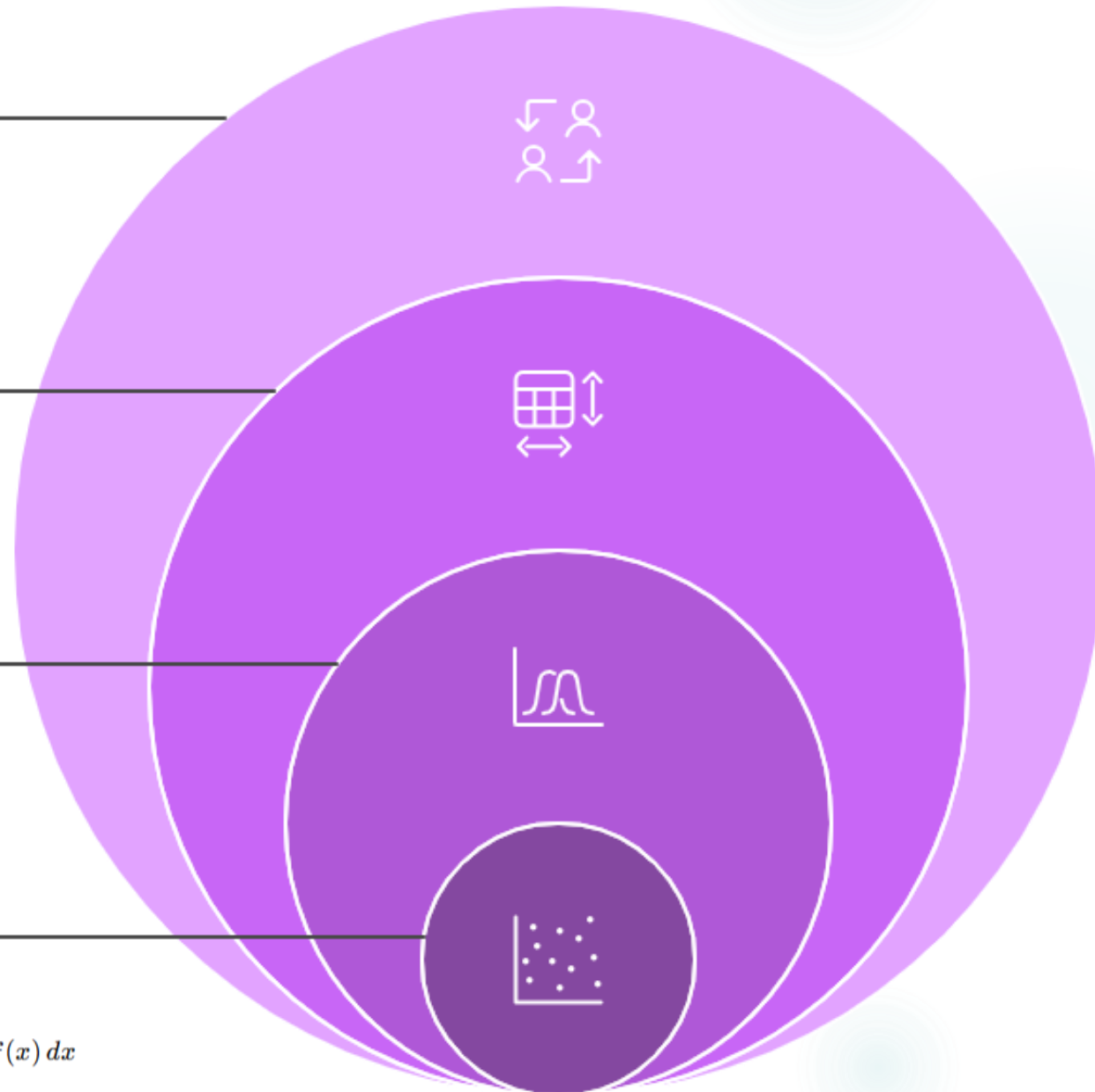
Spread of data from the mean

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

Mean

Central tendency of data
(Average)

$$\mathbb{E}[X] = \sum x_i P(x_i) \quad \text{or} \quad \int x f(x) dx$$



Probability Distributions

A probability distribution is a mathematical function or table that describes the likelihood of different outcomes for a random variable. In simpler terms, it shows how probabilities are spread across all possible values that a random variable can take.

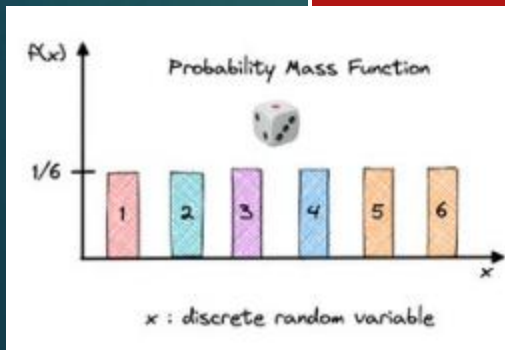
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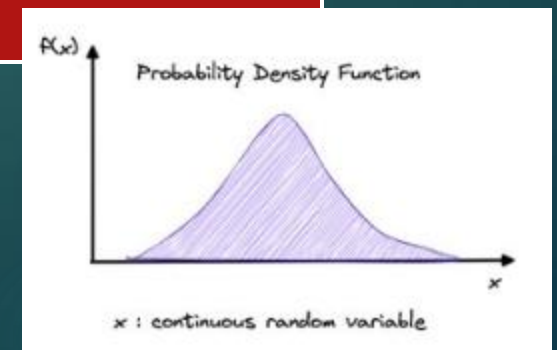
Probability
Mass Function
- PMF

Discrete

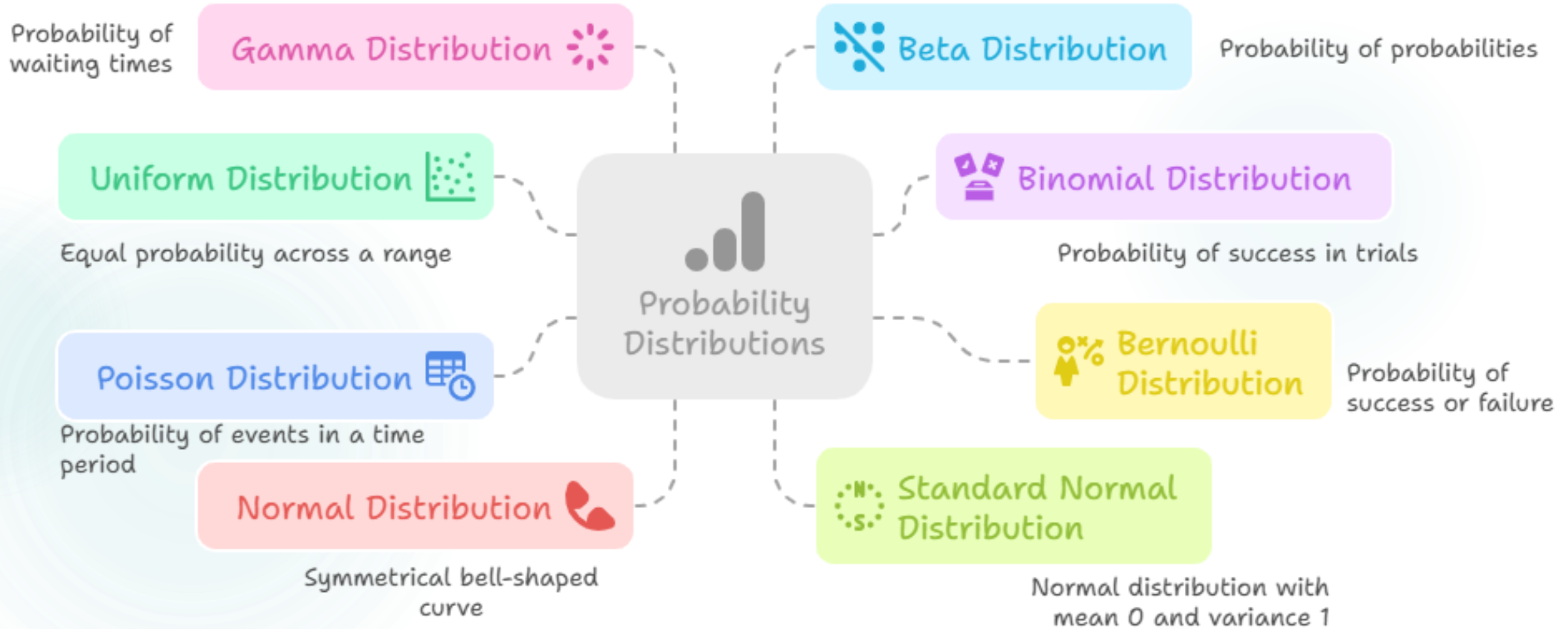


Probability Density
Function - PDF

Continuous

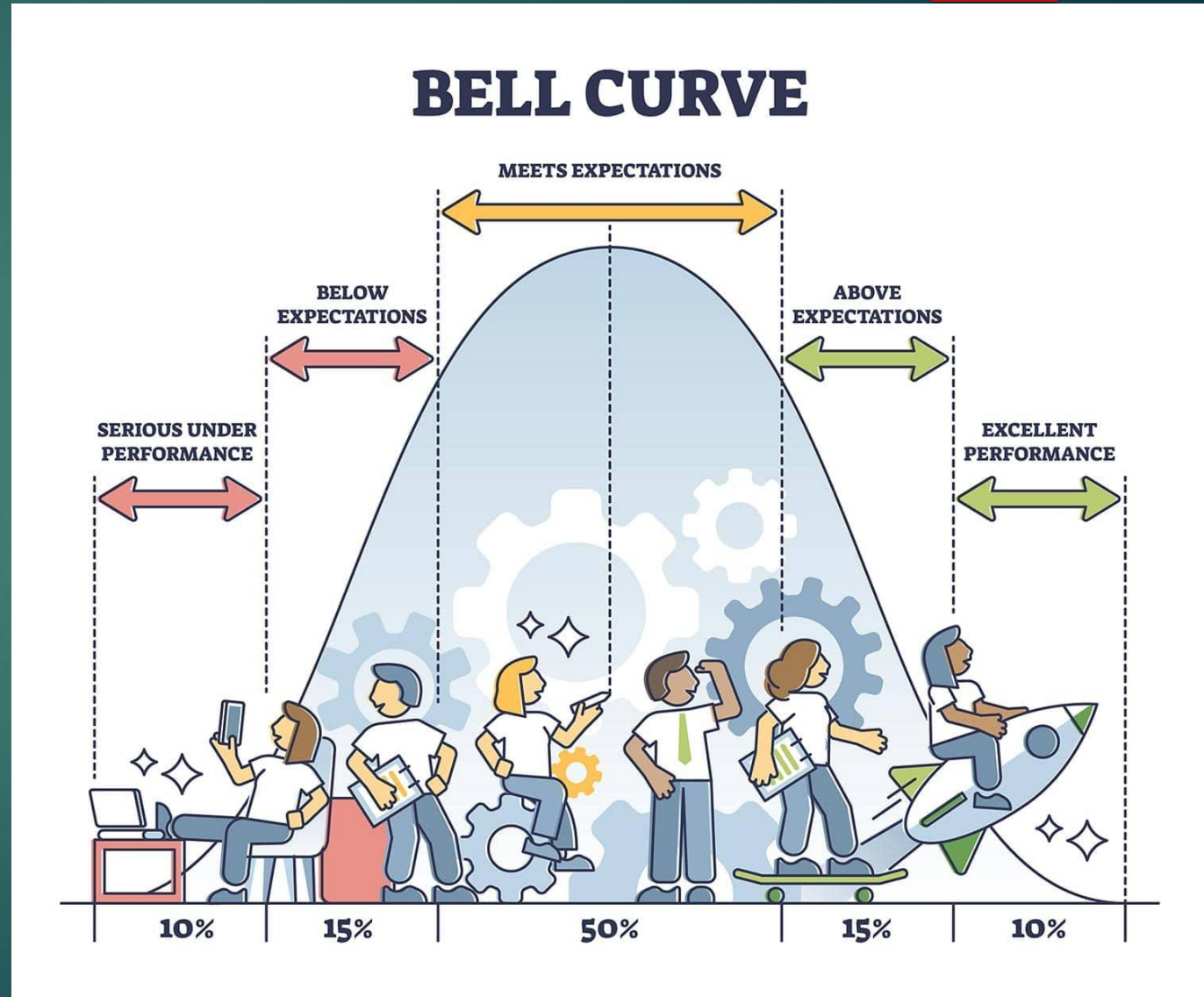


Probability Distributions Overview



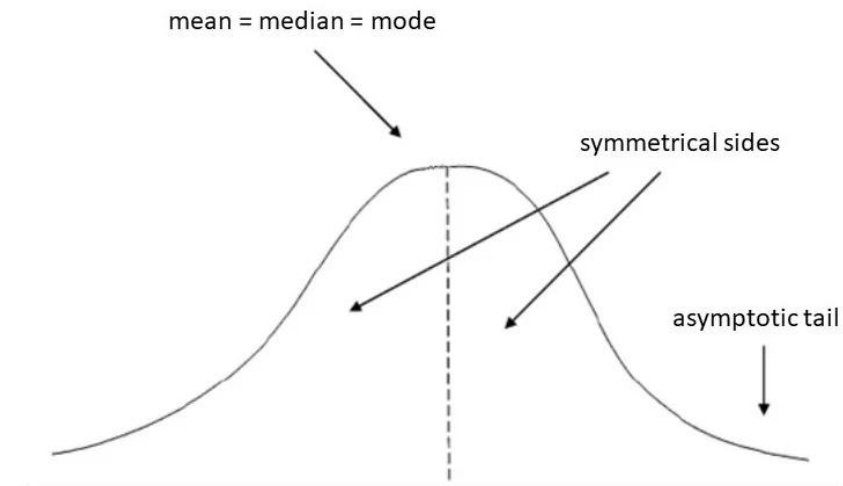
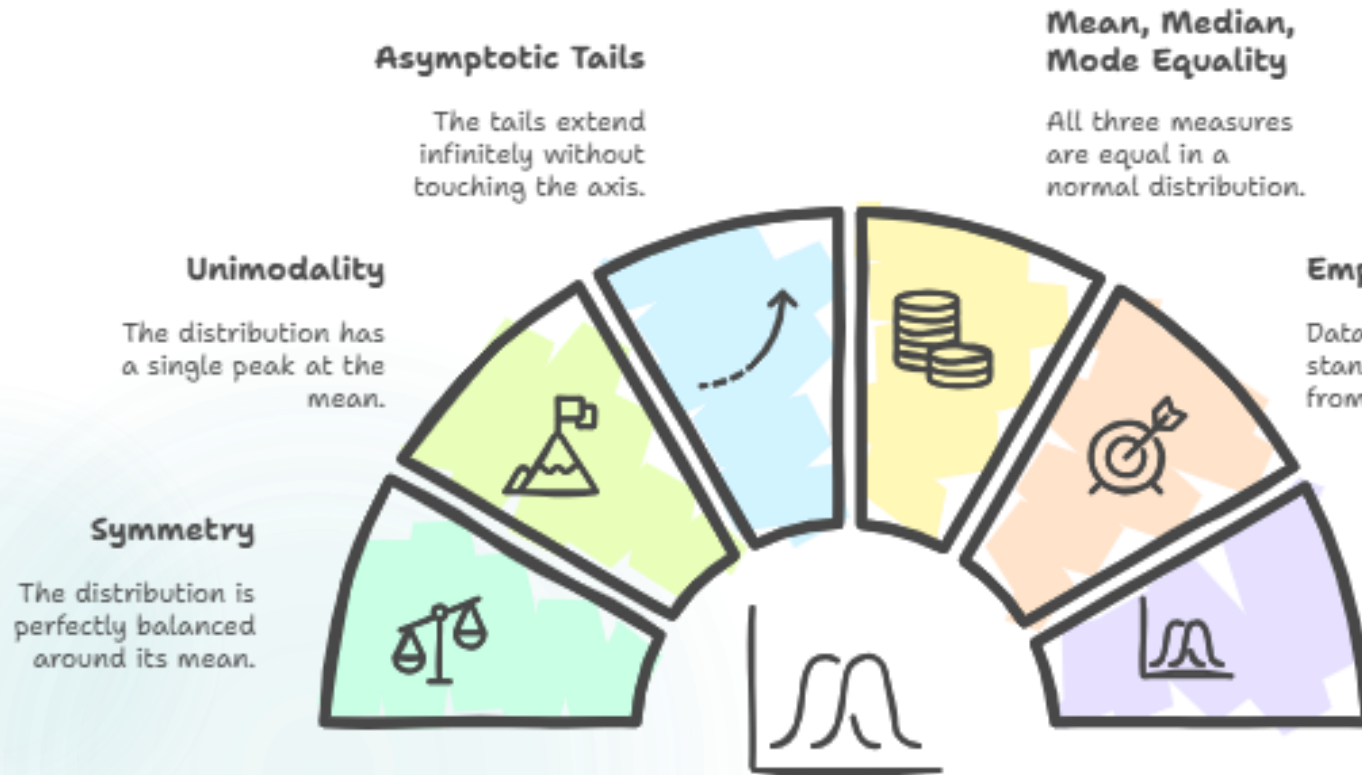
Normal Distribution

The normal distribution, also known as the Gaussian distribution, is a continuous probability distribution that is symmetrical around its mean. It is characterized by its bell-shaped curve, with the highest point at the mean and tapering off symmetrically on both sides.



Normal Distribution

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



Properties of Normal Distribution

1. The curve is bell-shaped and symmetric about the mean.
2. Mean = Median = Mode.
3. Total area under the curve is 1.
4. Defined by two parameters: mean (μ) and variance (σ^2).
5. 68% of data lies within 1σ from the mean.
6. 95% of data lies within 2σ from the mean.
7. 99.7% of data lies within 3σ from the mean.
8. The domain is from $-\infty$ to $+\infty$.
9. The curve approaches, but never touches, the x-axis.
10. Any linear combination of normal variables is also normally distributed.

A random variable X follows a normal distribution then it is denoted as
 $X \sim N(\mu, \sigma^2)$

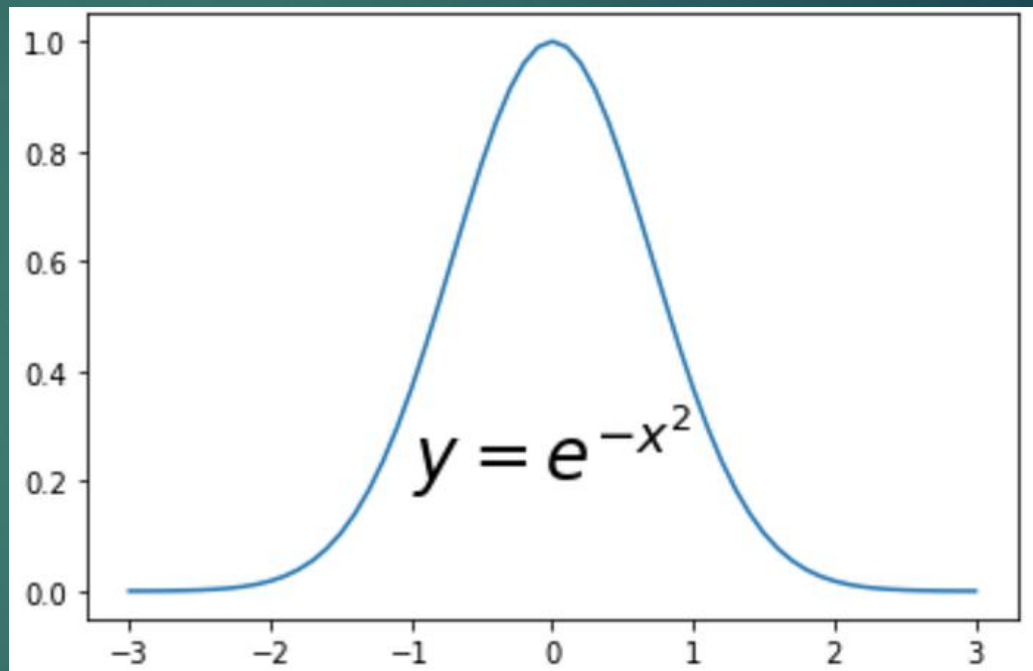
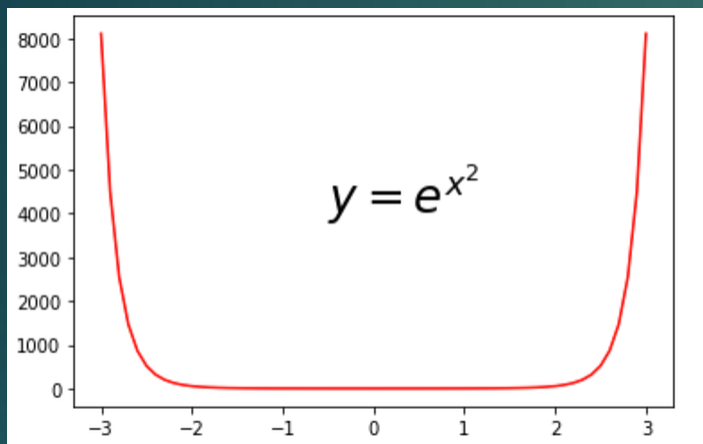
Probability Density Formula for Normal Distribution

$$f(x, \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x - \mu)^2}{2\sigma^2}}$$

x : Random Variable

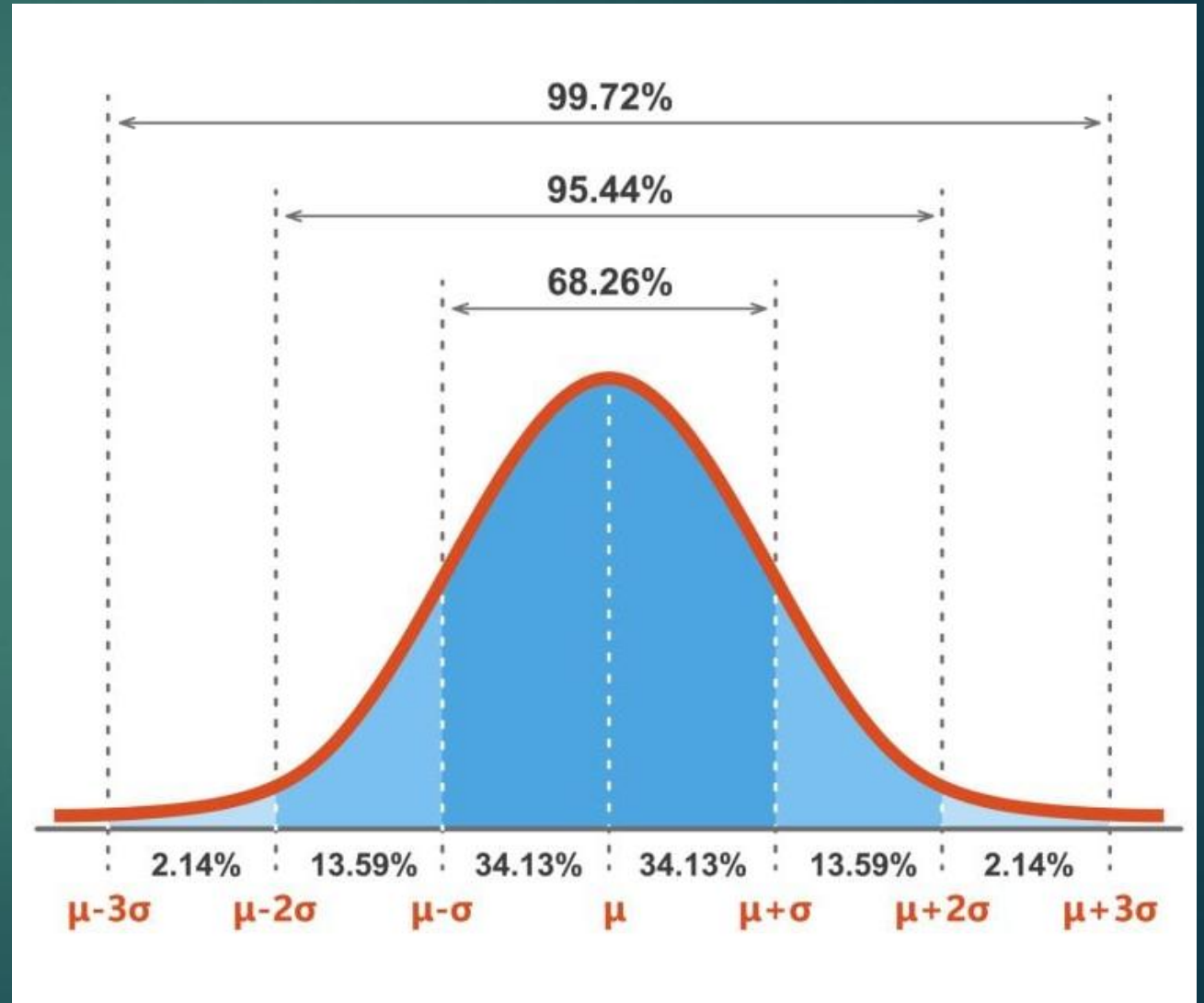
μ : Mean

σ : Standard Deviation



The empirical rule, or 68-95-99.7 rule

- ▶ Approximately 68% of the data falls within one standard deviation of the mean (z-scores between -1 and 1).
- ▶ Around 95% of the data falls within two standard deviations of the mean (z-scores between -2 and 2).
- ▶ Nearly 99.7% of the data falls within three standard deviations of the mean (z-scores between -3 and 3).

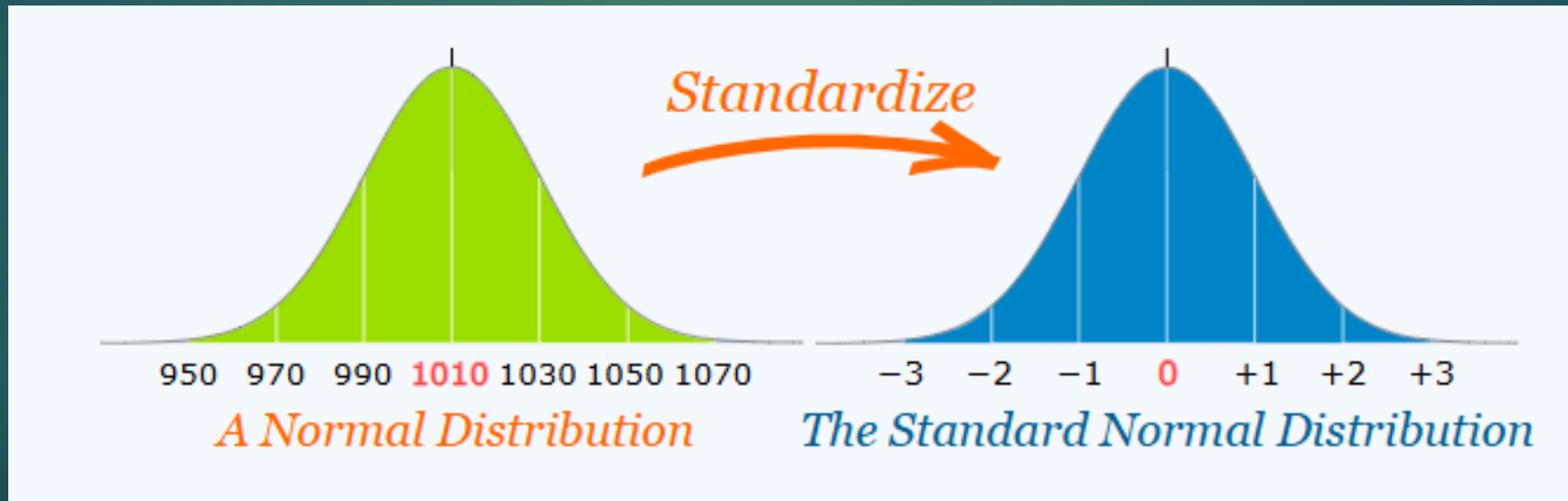


Standard Normal Distribution

A standard normal distribution is defined as a distribution with a mean of 0 and a standard deviation of 1.

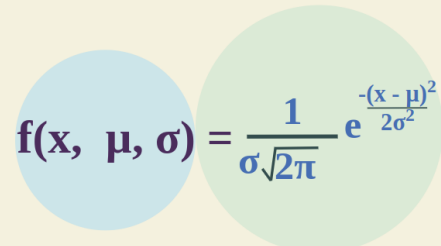
$$Z = \frac{x - \mu}{\sigma}$$

Score x Mean μ
SD σ



A random variable Z follows a standard normal distribution then it is denoted as $Z \sim N(0, 1)$

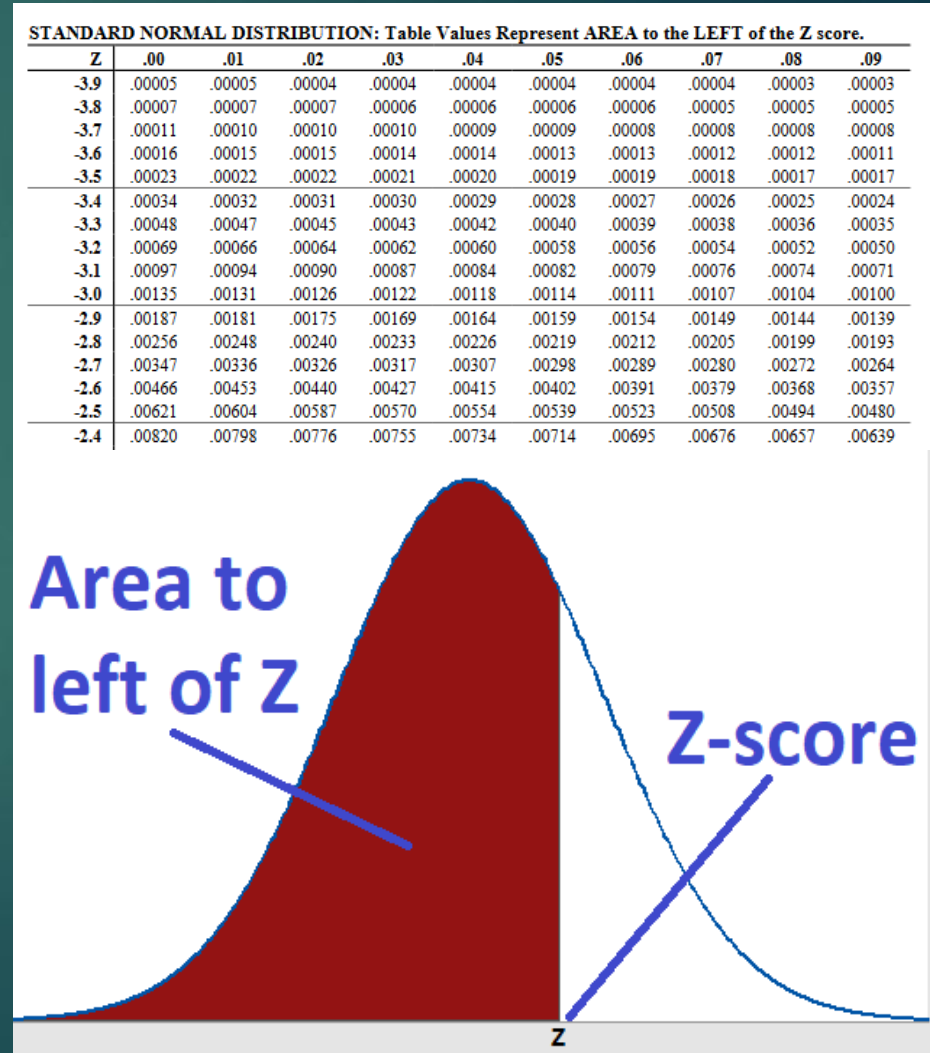
**Probability Density Formula for
Normal Distribution**


$$f(x, \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- PDF of Z : $f(z) =$
- Mean of Z : $E(Z) =$
- Variance of Z : $\text{var}(Z) =$

Standard Normal Distribution Table: Areas to the Left of the Z-Score

- The table you provided gives the area under the standard normal curve to the left of specific z-scores, which can be useful for finding the probability of a given data point falling below or above the mean.
- For example, the area to the left of a z-score of 1.00 is approximately 0.8413, which means that there is an 84.13% chance that a randomly selected data point from a standard normal distribution will fall below a z-score of 1.00.



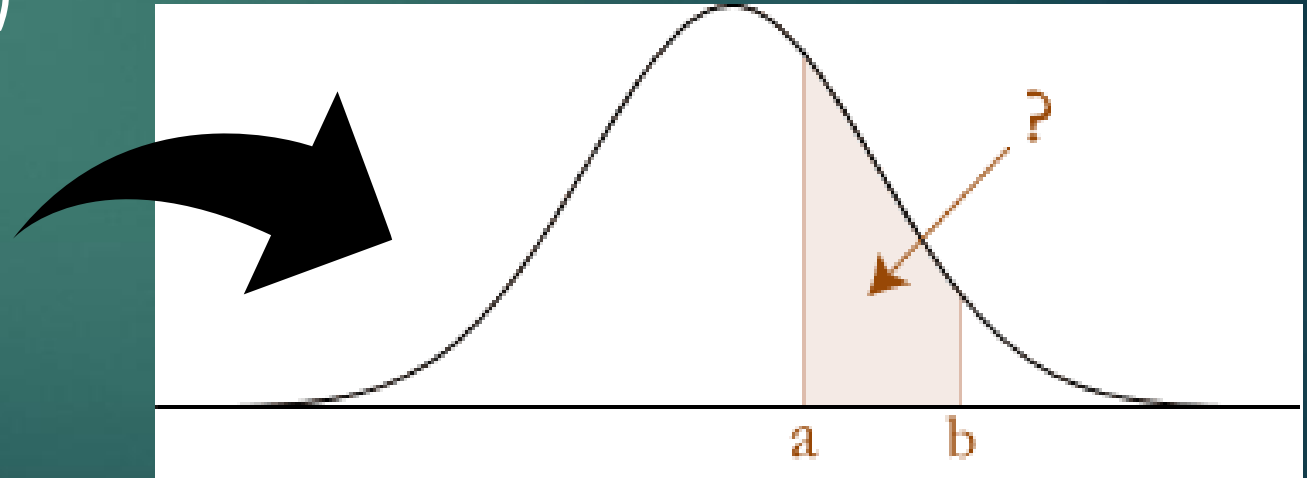
How to do Normal Distributions Calculations

$$P(a < X < b) \quad \text{where } X \sim N(\mu, \sigma^2)$$

$$P(a - \mu < X - \mu < b - \mu)$$

$$P\left(\frac{a - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{b - \mu}{\sigma}\right)$$

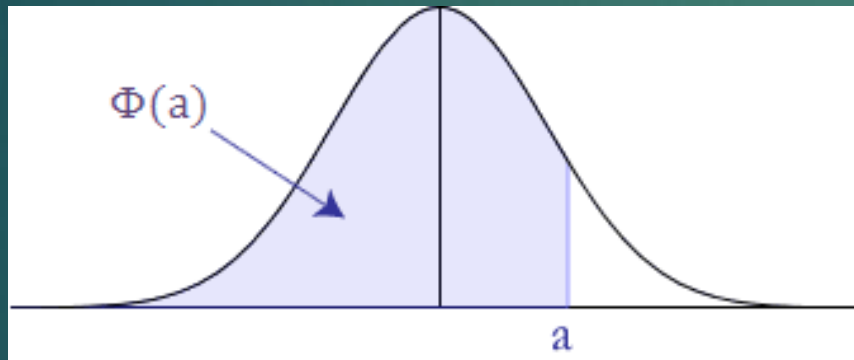
$$P\left(\frac{a - \mu}{\sigma} < Z < \frac{b - \mu}{\sigma}\right)$$



How to do Normal Distributions Calculations

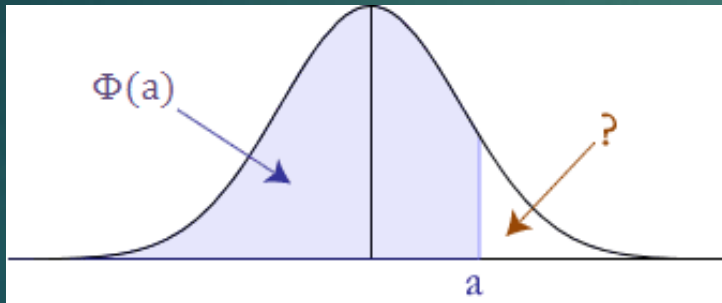
1) $P(Z < a)$

Probability less than a z-value



How to do Normal Distributions Calculations

2) $P(Z > a)$

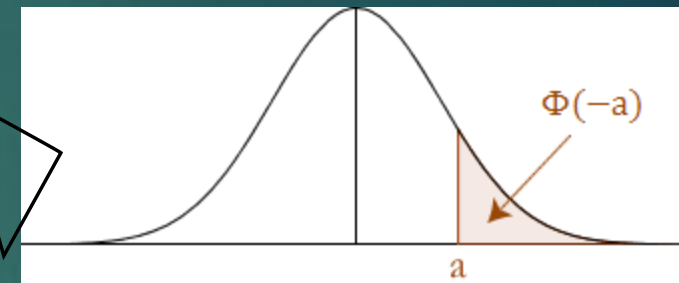
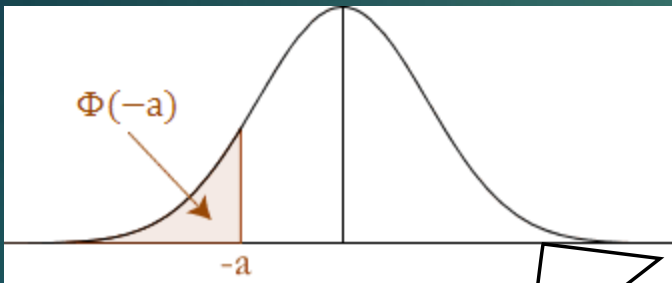


$$P(Z > a) + P(Z < a) = 1$$

$$P(Z > a) = 1 - P(Z < a)$$

How to do Normal Distributions Calculations

3) $P(Z < -a)$

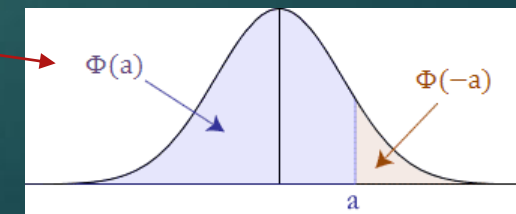


$$P(Z < -a) = P(Z > a)$$

$$P(Z > a) + P(Z < a) = 1$$

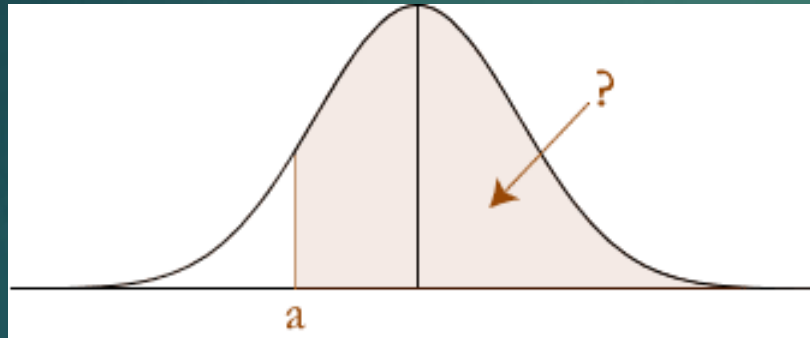
$$P(Z < -a) + P(Z < a) = 1$$

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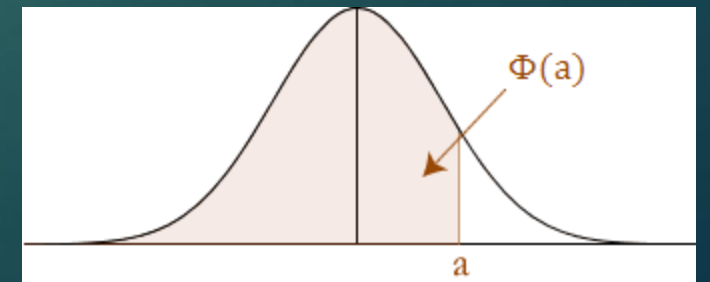


How to do Normal Distributions Calculations

4) $P(Z > -a)$

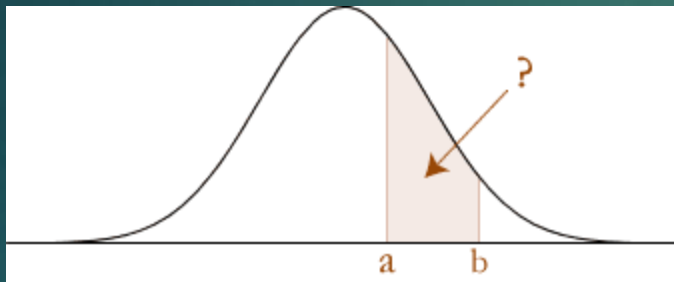


$$P(Z > -a) = P(Z < a)$$



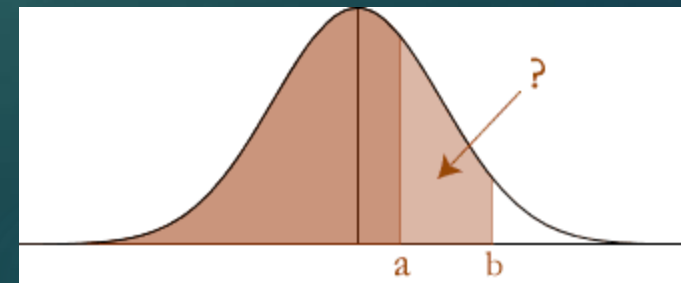
How to do Normal Distributions Calculations

5) $P(a < Z < b)$



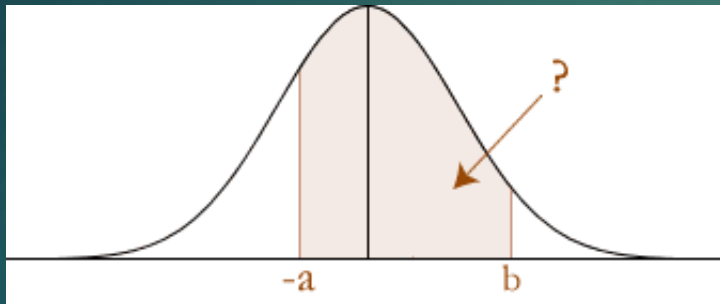
Probability between z-values

$$P(a < Z < b) = P(Z < b) - P(Z < a)$$



How to do Normal Distributions Calculations

6) $P(-a < Z < b)$

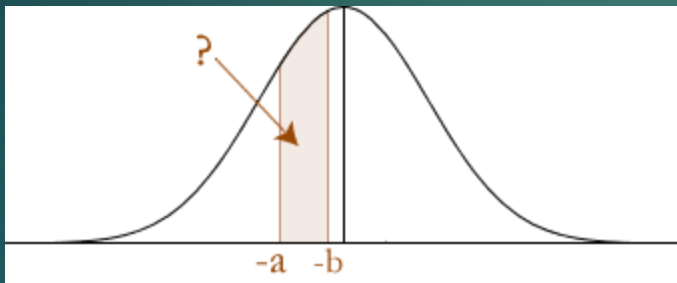


Probability between z-values

$$\begin{aligned} P(-a < Z < b) &= P(Z < b) - P(Z < -a) \\ &= P(Z < b) - (1 - P(Z < a)) \end{aligned}$$

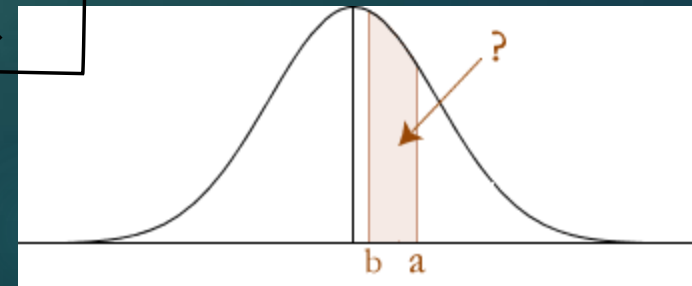
How to do Normal Distributions Calculations

7) $P(-a < Z < -b)$



Probability between z-values

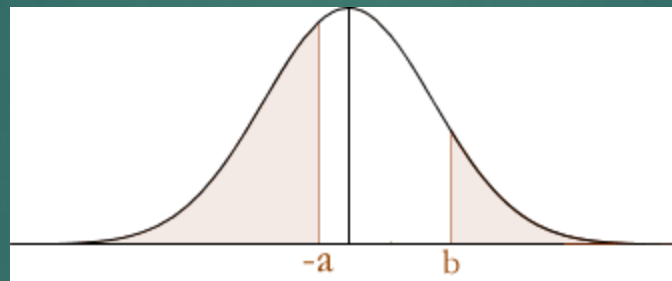
$$P(-a < Z < -b) = P(b < Z < a)$$



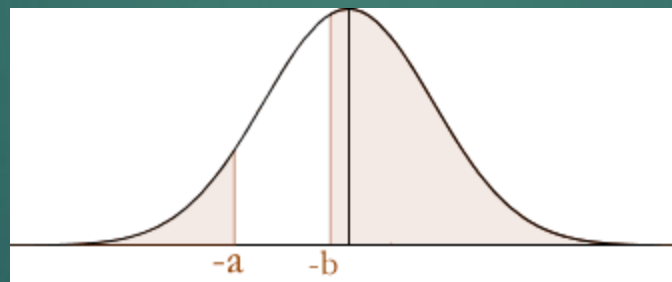
How to do Normal Distributions Calculations

Probability outside of a range of z-values

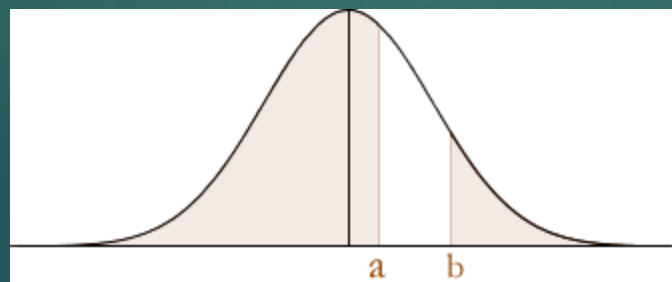
8) $P(Z < -a) + P(Z > b)$



9) $P(Z < -a) + P(Z > -b)$



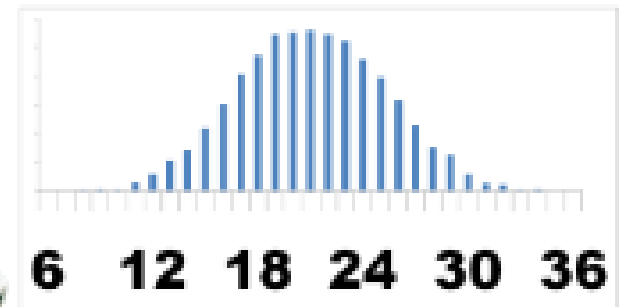
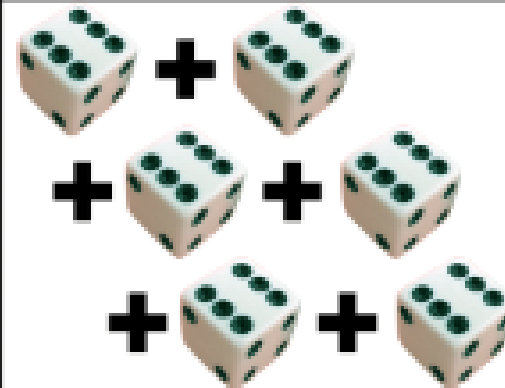
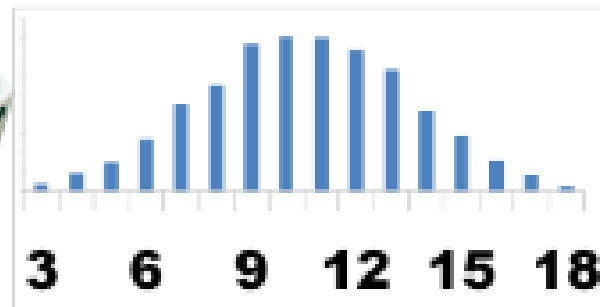
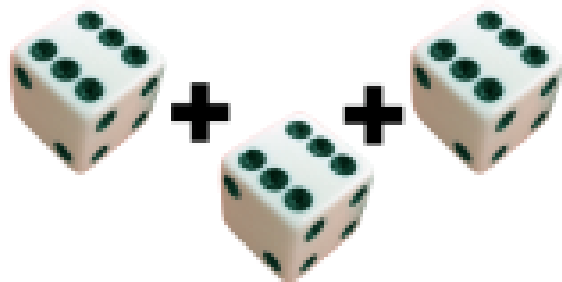
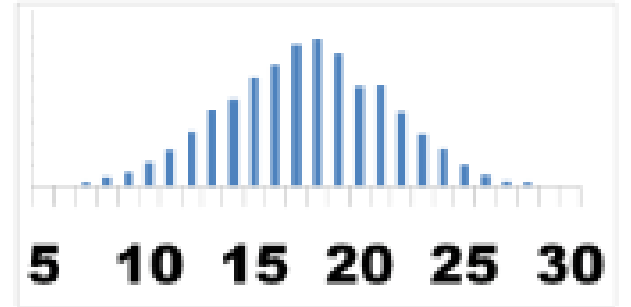
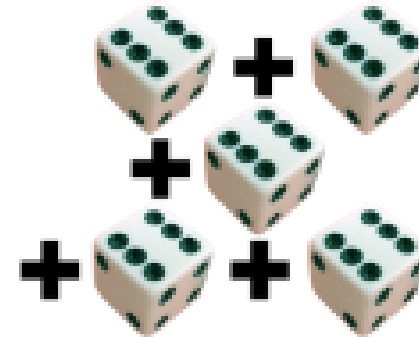
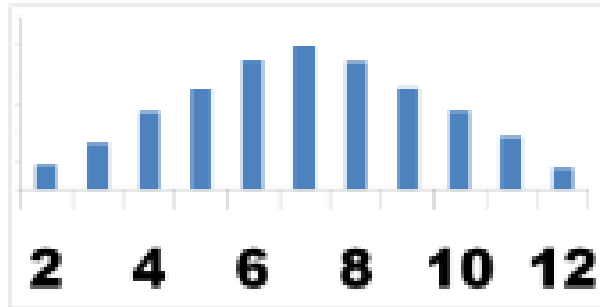
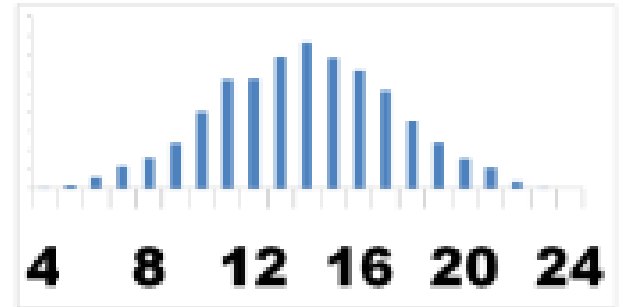
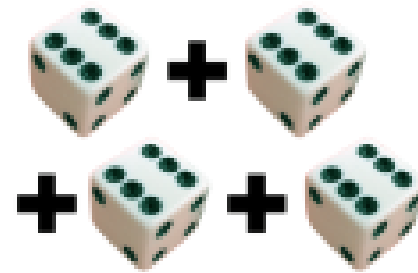
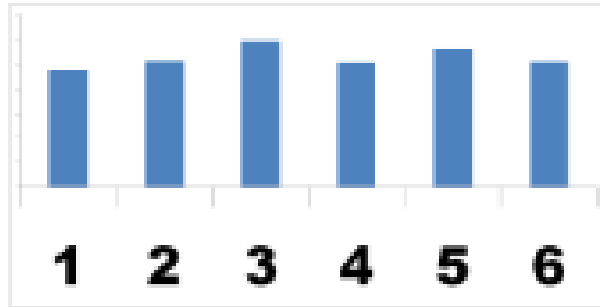
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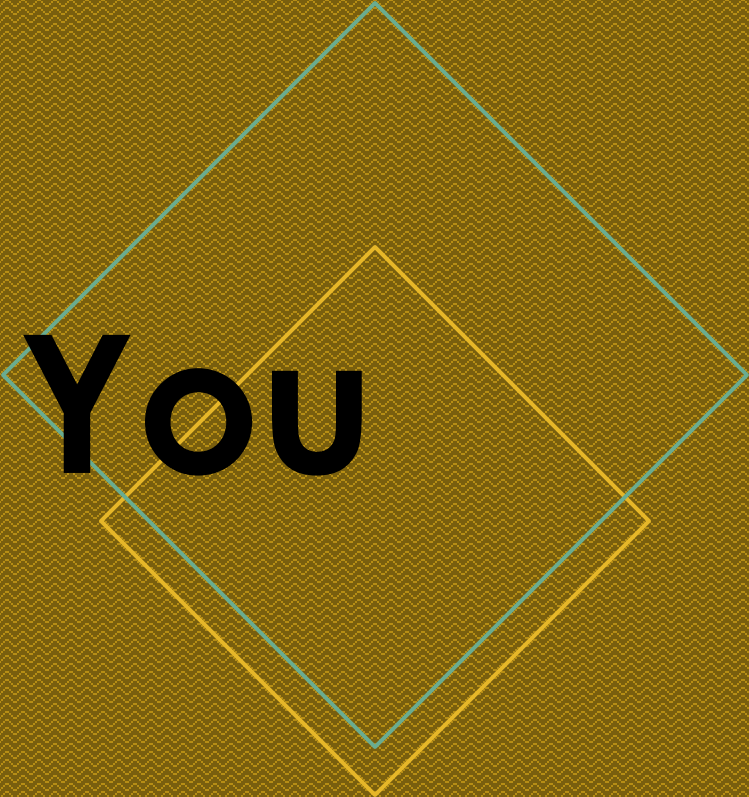


The Central Limit Theorem

- (Law of Large Numbers)

As the sample size approaches infinity, the probability distribution becomes a perfectly symmetrical (normal distribution) where the centre of the curve is the mean of the population, regardless of the original distribution of the population..





Thank You